

Blood Groups

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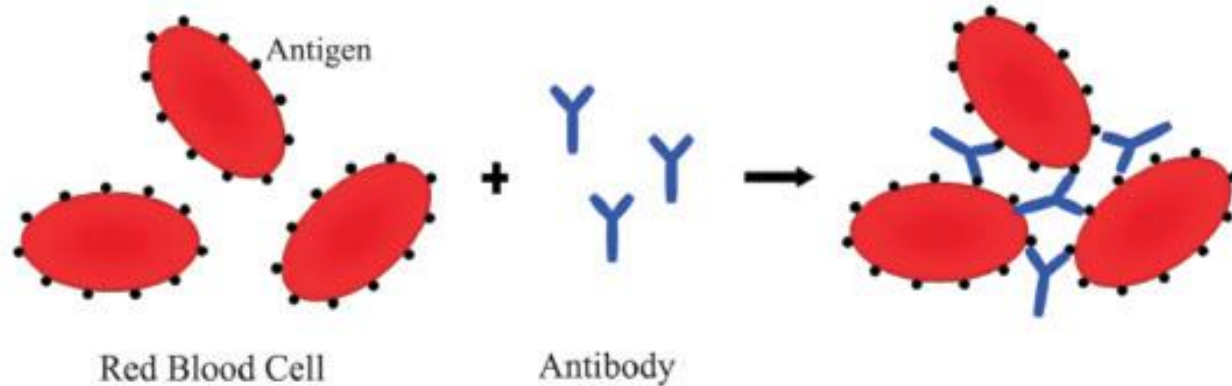
What is a Blood Group

Blood Groups

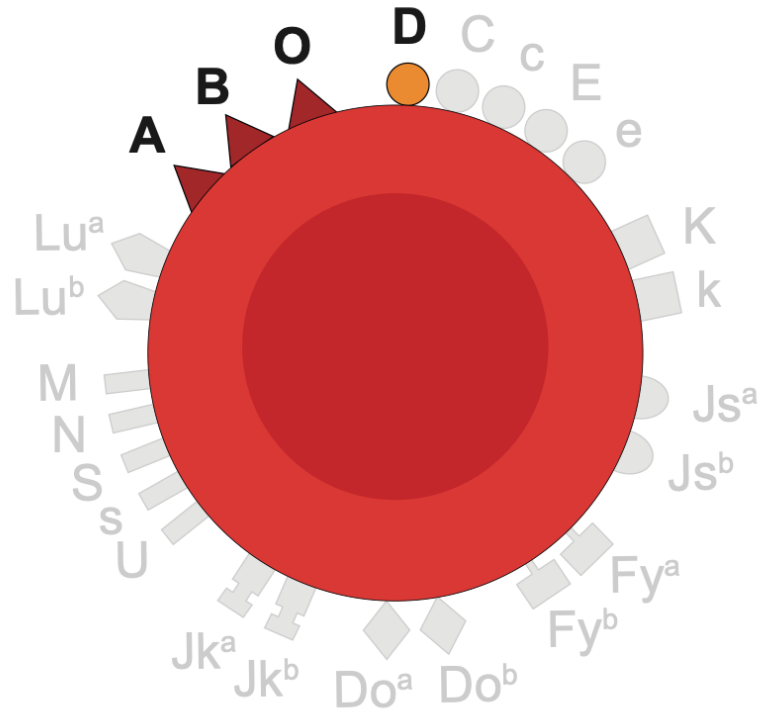
- Classification of blood type based on antigen presence
- Ex: ABO, Rh

Antigen

- Molecule that triggers an immune response
- Can be recognized by an antibody



RBC Antigens



Antibodies

Developing Antibodies

- Naturally occurring
 - Ex: ABO antibodies
- Can form after exposure
 - Transfusion
 - Pregnancy
 - Needle sticks

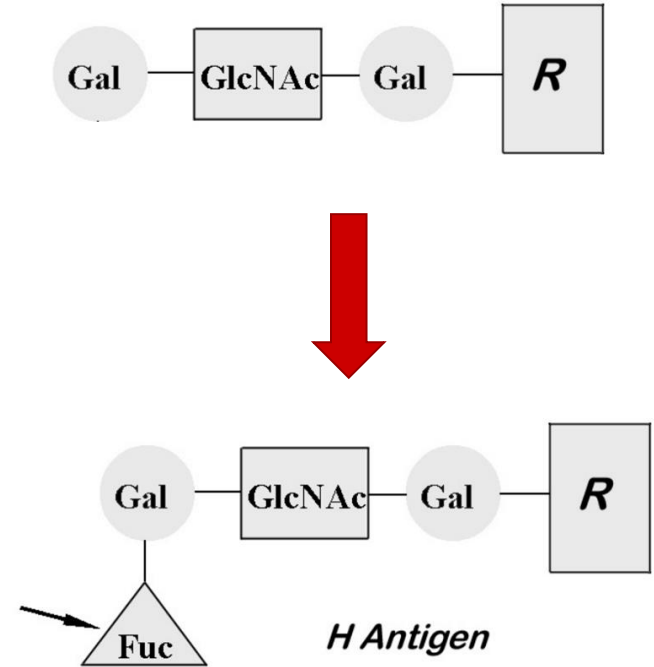
Antibody Problems

- Hemolytic transfusion reaction
 - Antigen-antibody interaction leads to destruction of RBCs
- Hemolytic disease of the fetus and newborn (HDFN)
 - Mother's antibodies attack and destroy the RBCs of fetus or neonate

ABO System

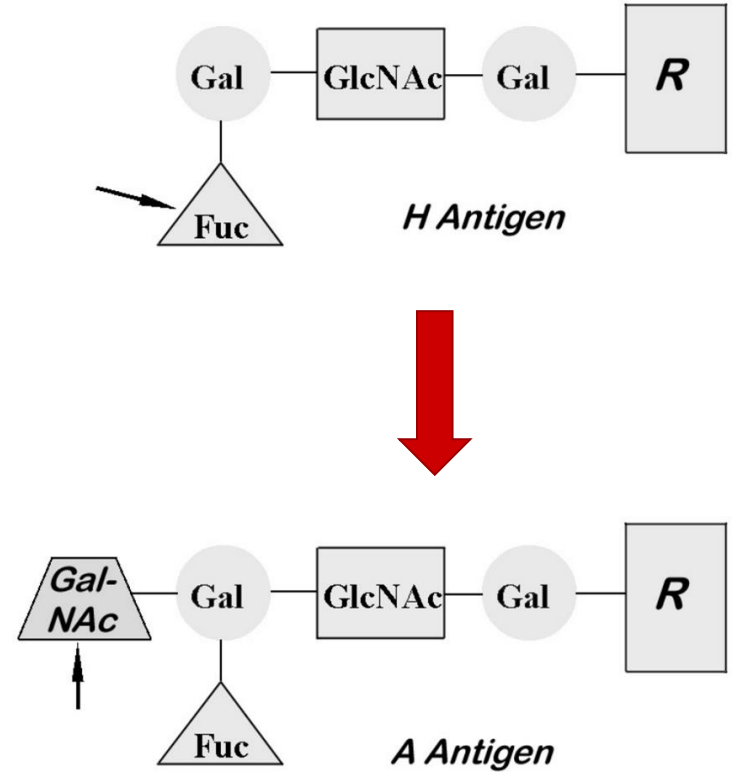
H Antigen

- H (FUT1) encodes fucosyltransferase
- Fucosyltransferase adds fucose to chain
- Building block for A and B antigens



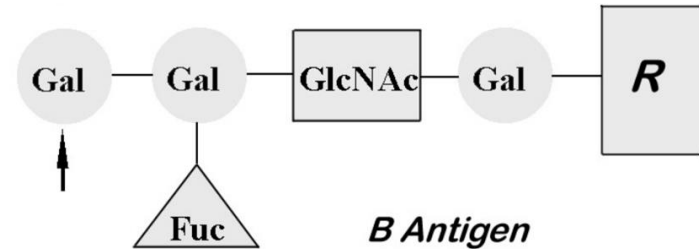
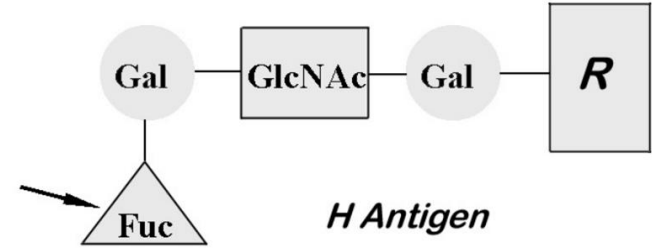
A Antigen

- Addition of N-acetylgalactosamine to H antigen



B Antigen

- Addition of Galactose to H antigen



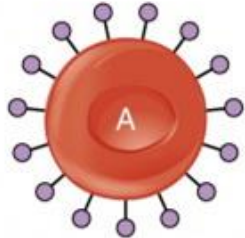
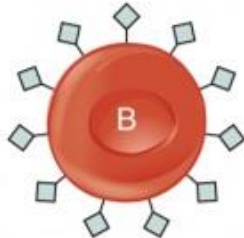
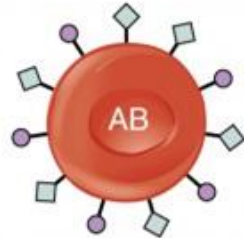
ABO Antigens

- Genes on chromosome 9 encode transferase enzymes for A and B
 - O is non-functional
- Also present on other cells

ABO Inheritance

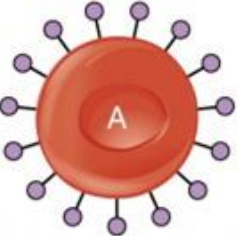
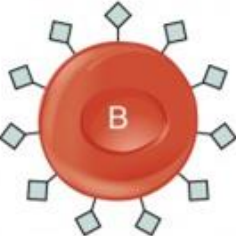
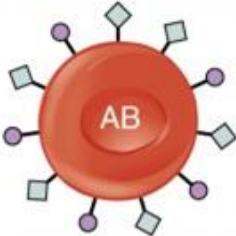
ABO genotype in the offspring		ABO alleles inherited from the mother		
		A	B	O
ABO alleles inherited from the father	A	A	AB	A
	B	AB	B	B
	O	A	B	O

ABO Blood Groups

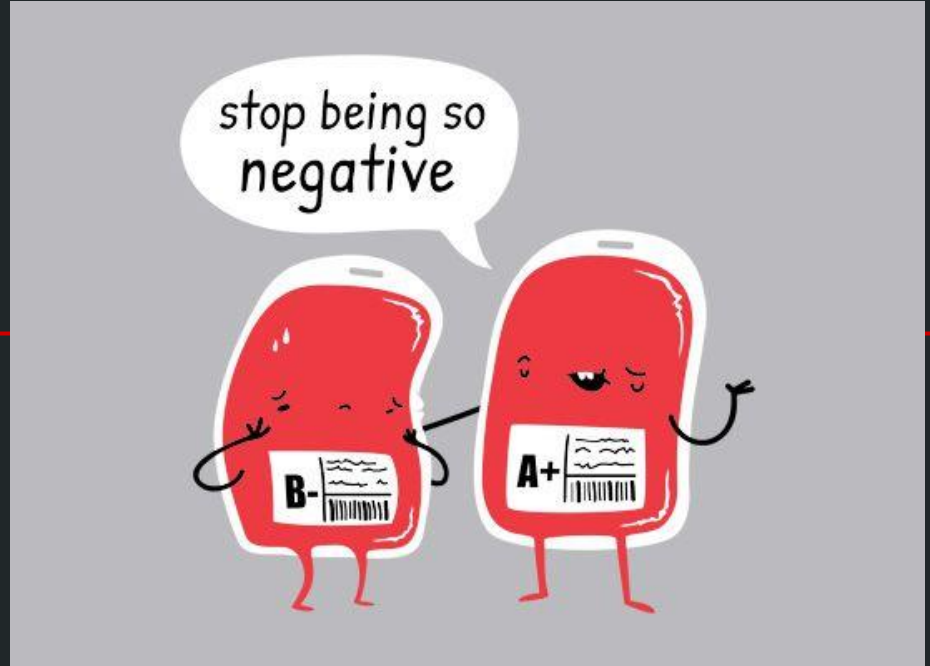
Blood Type				
	A	B	AB	O
Red Blood Cell Type				

ABO Antibodies

- Clinically significant
- Naturally occurring

		Blood Type			
		A	B	AB	O
Red Blood Cell Type					
Antibodies in Plasma		 Anti-B	 Anti-A	None	 Anti-A and Anti-B

Rh System



Antigens

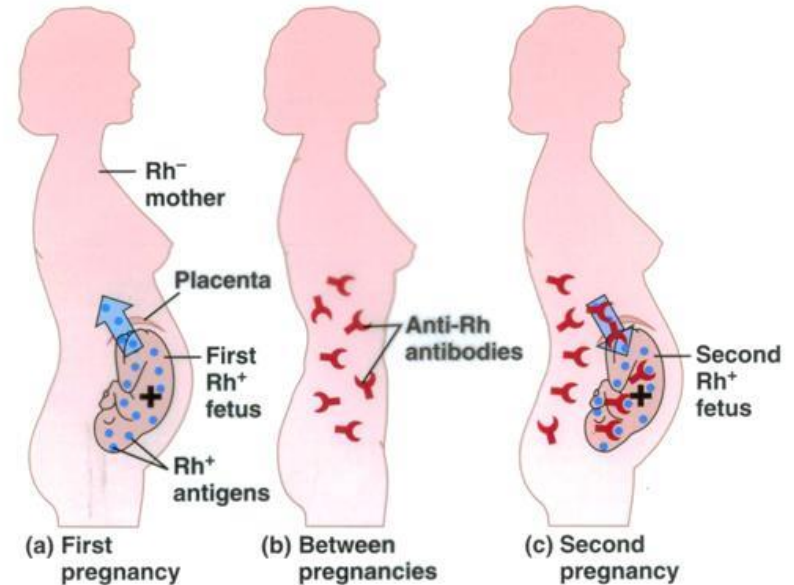
- D - “Rh positive”
- C
- E
- c
- e

Antibodies

- Require exposure
- D results in the most antibodies
- Can result in severe HDFN

Rhogam

- For Rh-negative moms who may have Rh-positive babies
- IgG anti-D
- Prevents creation of anti-D by mom



Other Blood Systems

Other Blood Systems

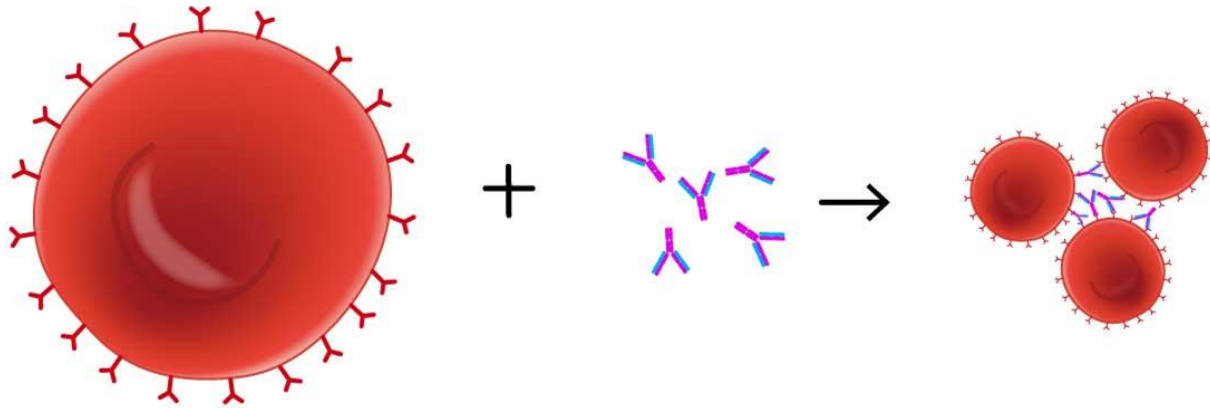
- Lewis
- I
- Kidd
- MNS
- Duffy
- Kell
- And many others!

Antibody Screen

	RH						MNS				LU		P	Lewis		Kell		Duffy		Kidd		LISS (Test Tube)		
	D	C	E	c	e	f	M	N	S	s	Lu ^a	Lu ^b	P ₁	Le ^a	Le ^b	K	k	Fy ^a	Fy ^b	Jk ^a	Jk ^b	IS	37 C	IAT
1	+	+	0	0	+	0	+	+	+	+	0	+	+	+	0	+	+	+	0	0	+	0	0	2+
2	+	0	+	+	0	0	+	0	+	0	0	+	+	0	+	0	+	+	+	+	0	0	0	1+
3	0	0	0	+	+	+	0	+	0	+	0	+	0	0	+	0	+	0	+	+	+	0	0	0✓

How to Test for ABO

Antibodies & Agglutination

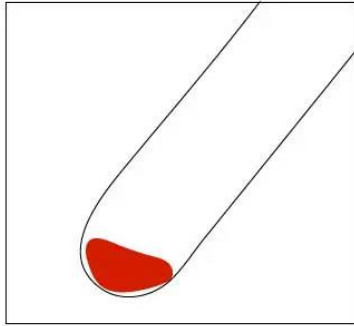


Types of Testing

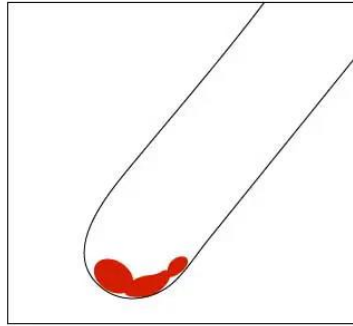
Tube Testing

Gel (Solid Phase) testing

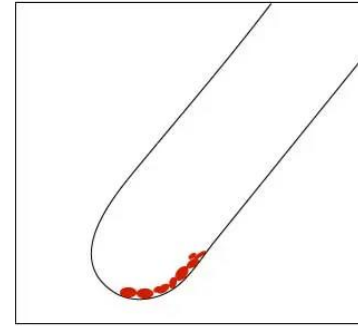
Tube Testing



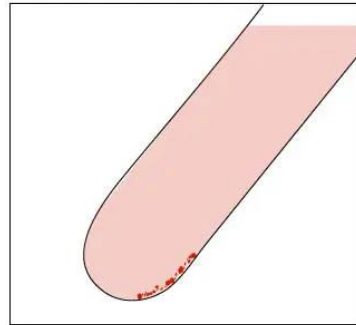
4+ Agglutination



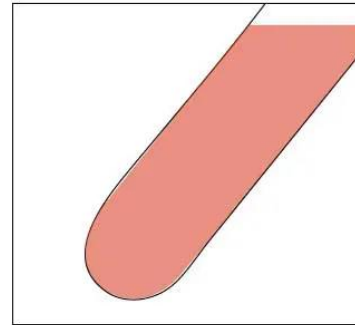
3+ Agglutination



2+ Agglutination



1+ Agglutination



Negative

Cell - Forward Type

- Use anti-A and anti-B with patient RBCs
- If A is present, cells will agglutinate with anti-A
- If B is present, cells will agglutinate with anti-B

Serum - Reverse/Back Type

- Use A and B RBCs with patient serum
- If patient serum has anti-A, A RBCs with agglutinate
- If patient serum has anti-B, B RBCs with agglutinate

Compare Forward & Reverse

Cell		Serum		ABO Group
Anti-A	Anti-B	A ₁ cells	B cells	
4+	0	0	4+	
0	4+	4+	0	
4+	4+	0	0	
0	0	4+	4+	